

Count Sort

I/P

A	4	9	2	4	7	6
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Step 1

Determine max value :- 9

Step 2

Create empty index array of size range

1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0

Step 3

fill index array with the number of occurrences

1	2	3	4	5	6	7	8	9
0	1	0	2	0	1	1	0	1

Step 4

Sum up index array value
with predecessor values.

1	2	3	4	5	6	7	8	9
0	1	1	3	3	4	5	5	6

Step 5 :-

Create an empty array to
store sorted values.

Input

4	9	2	4	7	6
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1	2	3	4	5	6	7	8	9
0	1	1	3	3	4	5	5	6

After every mapping, decrement the predecessor value by 1

1	2	3	4	5	6	7	8	9
0	0	1	2	3	3	4	5	5

decr by 1
↓

output

2	4	4	6	7	9
1	2	3	4	5	6

Time Complexity is $O(n)$

Pseudocode

Count Sort (array, n)

max = find max element in the
given array

Create Count array with
size maximum + 1

Initialize Count array with all
0's

for $i = 0$ to n

find the count of every
unique element and

store that count at i th
position in the count
array

for $j = 1$ to $\max$
Now, find the cumulative
sum and store it in
Count array

for $i = n$ to 1

Restore the array elements
Decrease the count of every
restored element by 1

end Count sort.